Online Supplementary Appendix

Mapping O*NET Tasks to ISCO Occupations using Text Similarity

Majid Einian

A Validation framework

Validation compares occupation-level assignments implied by the task-based mapping to institutional SOC–ISCO concordances constructed by statistical agencies and international organisations. Task-level assignments are first aggregated to the SOC occupation level. Each SOC occupation is assigned the ISCO-08 unit group receiving the largest number of task matches. Agreement is then evaluated relative to institutional SOC–ISCO mappings at both the four-digit unit-group level and the one-digit major-group level.

Multiple institutional concordances are used in order to avoid dependence on any single mapping source. Agreement is therefore evaluated relative to both strict and lenient benchmarks. The strict benchmark requires agreement with all available crosswalks for a given SOC occupation, while the lenient benchmark requires agreement with at least one institutional source. Results are similar across benchmark definitions, indicating that agreement rates are not driven by any single concordance.

Institutional crosswalks are used only for validation and do not influence construction of the task-level mapping.

B Full validation results

Table B1 reports validation results for alternative benchmark definitions and O*NET releases. The main text reports results for O*NET 29.2 using the lenient benchmark, which balances measurement error across institutional concordances.

Table B1: Chain crosswalk validation: match rates across scenarios.

Scenario	Cov. (%)	Exact (%)	Sub-major (%)	Major group (%)
*SOC 2018 (O*NET 29.2)*				
A1 – ESCO–SOC 2018	99.9	42.7	62.3	77.5
A2 – SOC 2018–ESCO	98.7	35.8	54.1	67.9
A3 – Strict intersection	98.5	34.9	53.5	67.7
A4 – Lenient union	99.9	43.6	63.0	77.7

Continued on next page

Table B1 – continued

Scenario	Cov. (%)	Exact (%)	Sub-major (%)	Major group (%)
*SOC 2010 (O*NET 25.0)*				
B1 – ESCO–O*NET (Matysiak et al.)[a]	79.1	26.3	43.7	58.6
B2 – SOC 2010–ISCO-08 (BLS)	99.9	20.3	37.5	50.7
B3 – Lenient union (Matysiak $\cup$ BLS)	100.0	25.5	44.4	58.5
B4 – Strict intersection (Matysiak $\cap$ BLS)	70.0	22.4	37.8	50.4

[a]Less independent: derived via the same semantic similarity approach as the pipeline.

C Author annotation of O*NET 29.2 task assignments

To complement the automated chain-crosswalk validation, 108 O*NET 29.2 tasks were selected and annotated by the author. Tasks were drawn from 36 SOC occupations covering all nine ISCO major groups, with occupations chosen in order of combined US and Nordic employment within each group and tasks selected by O*NET importance score (top three per occupation). For each task the author assigned the most appropriate ISCO-08 four-digit unit group from a shortlist of plausible candidates, without reference to the pipeline's predictions.

The selected configuration ($w_{\text{isco}} = 0.828$, $w_{\text{dwa}} = 0.037$, $w_{\text{soc}} = 0.599$, max links$= 1$; see Appendix D for the parameter selection procedure) achieves 61.1% exact four-digit agreement with the author annotations, rising to 76.9% at the two-digit sub-major level and 81.5% at the one-digit major-group level, at a mean cosine similarity of 0.704. These results confirm that the semantic similarity approach produces assignments consistent with expert judgement at the occupation-group level.

D Parameter selection robustness

Parameter values are selected through a two-phase unsupervised search over blend weights and similarity thresholds governing construction of task and occupation representations.

Search procedure

Phase 1 (initial random sweep). 500 configurations are drawn by Latin hypercube sampling from a broad parameter space (seed 42), labelled `ONET29_sw0000`–`sw0499`. Parameters varied include w_{soc}, w_{isco}, w_{dwa}, the minimum cosine similarity threshold, the margin between the best and second-best match, and overload-control parameters.

Phase 2 (focused search). A fine 2D grid of 189 configurations crosses seven values of w_{dwa} (0.00–0.30) with nine values of w_{soc} (0.30–0.70) at three ISCO-weighting settings ($w_{\mathrm{isco}} \in \{0.0, 0.4, 0.8\}$), labelled `fg_isco##_dwa##_soc##`. An additional 500 configurations are drawn by targeted random sampling of the high-performing region identified in Phase 1 (seed 99), labelled `fr0000`–`fr0499`.

The two phases yield 689 candidate specifications in total. All variants in both phases share pre-computed Tier-1 embeddings (checkpoint prefix `ONET29`), so the variable cost is the fast blend and retrieval step only.

Composite score

Each candidate is scored by a weighted average of nine normalised metrics (Table D1). Each metric is min–max normalised to $[0, 1]$ within the candidate set before aggregation.

Table D1: Composite score components.

Metric	Weight	Direction
ISCO-08 coverage share	3	maximize
Mean cosine similarity retained	2	maximize
Share of tasks in overloaded ISCO group	2	minimize
Gini coefficient of task count per ISCO group	2	minimize
Best-link agreement rate	2	maximize
Jaccard similarity of link sets	2	maximize
Low-confidence retrieval share	1	minimize
Retrieval gap (rank-1 minus rank-2 similarity)	1	maximize
Mean links per task	1	minimize

The selected configuration (`fr0043`) achieves the highest composite score among focused-sweep Pareto-optimal candidates. Pareto candidates are identified across the six primary objectives (coverage share, mean similarity, overload share, Gini coefficient, best-link agreement, Jaccard similarity).

Findings

Including Detailed Work Activity (DWA) labels in the task representation consistently reduces performance across evaluation criteria. DWA text largely duplicates vocabulary already present in the task statements and appears to dilute occupation-specific language. All reported results therefore use $w_{\mathrm{dwa}} = 0$.

Balanced weighting of task text and SOC occupation titles produces stable assignments across specifications. Similarly, combining ISCO-08 task descriptions with ESCO occupation information improves performance relative to using either source separately, indicating that the two sources provide complementary information.

Each task is assigned to a single ISCO-08 unit group. Allowing multiple assignments increases coverage but reduces interpretability and complicates aggregation of task-based measures. High-performing configurations cluster in a narrow region of the parameter space, indicating that results are robust to reasonable variation in parameter values.

E Coverage and example matches

The mapping assigns O*NET tasks to 435 of the 436 ISCO-08 unit groups. The only group without assigned tasks is ISCO 1113 (Traditional Chiefs and Heads of Village), which has no functional equivalent in O*NET.

Coverage is broad across major occupational groups, including professional, technical, clerical, service, and manual occupations. Differences in task counts across ISCO-08 groups reflect differences in the level of detail in O*NET task descriptions and structural differences between U.S. and international occupational classifications.

Table E1 reports a small set of illustrative task assignments.

Table E1: Examples of task assignments

O*NET task description	ISCO-08 group
Prepare financial reports and analyse budget performance	2411 Accountants
Write and test computer code for software applications	2512 Software developers
Assess patient symptoms and recommend treatment plans	2211 Generalist medical practitioners
Install electrical wiring and equipment	7411 Building electricians